

Event Discovery Web Application

Group 1 - Phase 3 API Documentation and Implementation

Members:

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I. API Used

The Spott website uses a PHP JSON API connected to a MySQL database through XAMPP. The browser communicates with PHP endpoint files in the `api/` directory using JavaScript `fetch()` requests. The database is named `spott_db`. PDO is used in `config/db.php` to establish the connection between the PHP API and MySQL.

II. API Purpose

The API provides the event-discovery data and actions required by the Spott web application. It connects the responsive frontend to the MySQL database and allows the website to display current event information without hard-coding the event cards in the HTML.

- Retrieve active local events and related event information.
- Search and filter events by keyword, category, city, date, and maximum price.
- Load event categories for the category filter.
- Save event registrations for the demo attendee.
- Record confirmation that an event is still happening.
- Submit reports about potentially incorrect event listings.

III. API Endpoints

- GET - `api/events.php` - Returns active events and applies search/filter parameters.
- GET - `api/categories.php` - Returns the available event categories.
- POST - `api/register.php` - Registers the demo attendee for an event.
- POST - `api/confirm.php` - Updates the event still-happening confirmation timestamp.
- POST - `api/report.php` - Creates an event report for administrator review.

The events endpoint accepts the following query parameters:

- q - searches event titles, descriptions, organizers, venues, and cities.
- category - filters events by category name.
- city - filters events by city.
- date - filters events by event date.
- max_price - returns events at or below the selected price.

IV. Data Retrieved from the API

The api/events.php endpoint retrieves and combines data from the events, organizers, locations, categories, event_category, and registrations tables.

- Event details: event ID, title, description, start date/time, end date/time, price, and status.
- Organizer details: organization name and verification status.
- Location details: venue name, address, city, latitude, and longitude where available.
- Category details: one or more category names associated with the event.
- Engagement details: number of registered attendees.
- Trust details: timestamp showing whether the event was recently confirmed as still happening.

The api/categories.php endpoint retrieves category IDs and category names for the event filter dropdown.

Example response from api/events.php:

```
{
  "success": true,
  "count": 1,
  "events": [{
    "event_id": 1,
    "title": "Campus Music Night",
    "city": "Dasmariñas",
    "price": "150.00",
    "registrations": 0
  }]
}
```

V. How the API Is Integrated into the Website

The API is integrated through assets/app.js. The frontend creates requests based on the user's search and filter selections, sends them to the PHP endpoints, and uses the JSON responses to update the page dynamically.

1. The page loads and JavaScript requests categories from api/categories.php.
2. The category response is used to populate the category filter.
3. The page requests events from api/events.php.
4. The selected search and filter values are converted into URL query parameters.
5. The PHP endpoint uses PDO prepared statements to query MySQL.
6. The endpoint returns a JSON response containing the event records.

7. JavaScript renders the response as event cards, event details, counts, and map-location cards.
8. User actions such as RSVP, confirmation, and reporting send JSON POST requests to the corresponding endpoints.
9. The returned success or error message is displayed through the website notification system.

Example Fetch request:

```
const response = await fetch('api/events.php?' + params.toString());
const data = await response.json();
renderEvents(data.events);
```

The website does not reload the entire page when events are searched, filtered, registered, confirmed, or reported. This demonstrates AJAX/Fetch functionality required for Phase 3.

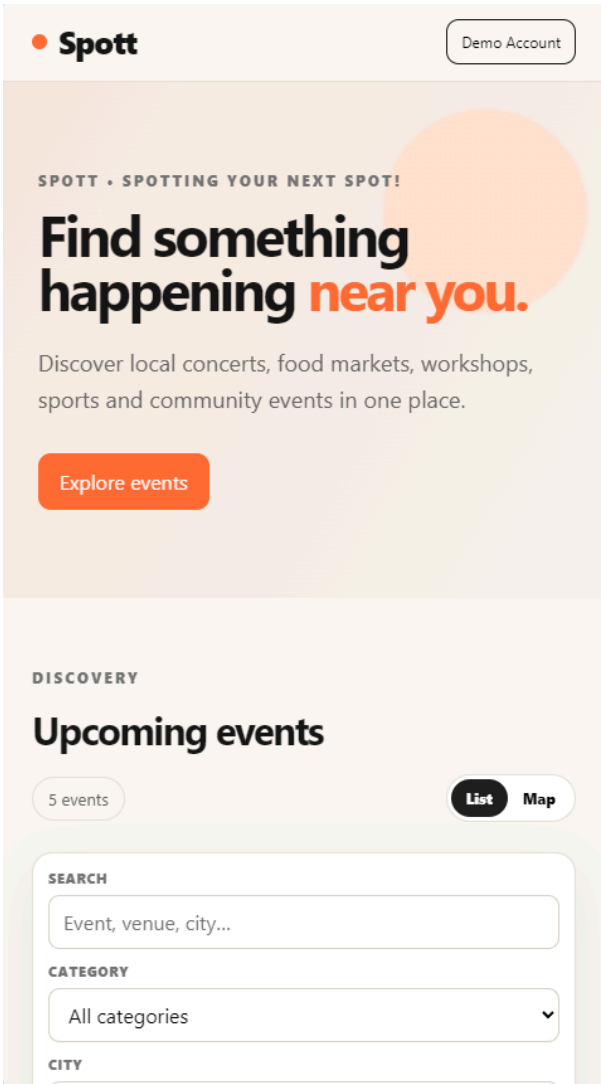
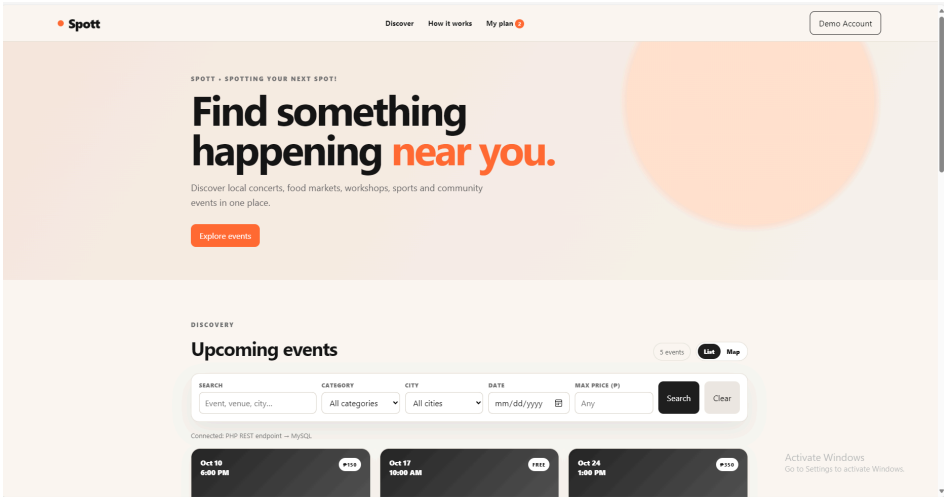
VI. Error Handling and Validation

- The API returns JSON success or error messages.
- Invalid event IDs are rejected by the PHP endpoints.
- Report submissions require a reason of at least five characters.
- The report field is limited to 500 characters.
- Database connection errors identify that XAMPP MySQL must be running.
- If no events match the filters, the website displays an appropriate empty-results message.

VII. Contribution of Each Member

Member	Role	Contribution	Rating
Marc Dei Niel B. Bides	UI/UX Design, QA Tester, Documentation	UI/UX wireframing, QA testing, and technical documentation.	10
Raphael Clarke A. Martinez	Frontend	Frontend development and responsive design implementation.	10
Khen Cyrus M. Rosal	Backend, Database	Backend architecture, database schema design, and API development.	10

VIII. Screenshots



Upcoming events

5 eventsListMap

SEARCH

Event, venue, city...

CATEGORY

All categories

CITY

All cities

DATE

mm/dd/yyyy

MAX PRICE (P)

Any

Search

Clear

Connected: PHP REST endpoint → MySQL

Oct 10
6:00 PM

P150

Music

Campus Music Night

Live performances and student bands.

DLSU-D Activity Center - Dasmariñas

1 registered - Still happening ✓

✓ Verified organizer

View details

Saved ✓

RSVP

Oct 17
10:00 AM

FREE

Community, Food

Cavite Food Weekend

Local food stalls and community vendors.

The District Imus - Imus

1 registered - Still happening ✓

✓ Verified organizer

View details

Saved ✓

RSVP

Oct 24
1:00 PM

P350

Workshop

Tagaytay Creative Workshop

A beginner-friendly creative workshop.

People's Park - Tagaytay

0 registered - Status not recently confirmed

✓ Verified organizer

View details

Save

RSVP

Nov 7
8:00 AM

P50

Community, Sports

Cavite Community Sports Day

Friendly games and community activities.

Cavite Provincial Capitol - Trece Martires

1 registered - Still happening ✓

✓ Verified organizer

View details

Saved ✓

RSVP

Nov 14
2:00 PM

FREE

Community, Workshop

Weekend Makers Meetup

Meet local makers, designers and hobbyists.

SM City Bacoor - Bacoor

1 registered - Status not recently confirmed

✓ Verified organizer

View details

Saved ✓

RSVP

Spott

DiscoverHow it worksMy plan

Demo Account

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Saved ✓

RSVP

Report this listing

If event information looks incorrect, send it to the administrator.

Reason for report (required)

Submit report

Spott

DiscoverHow it worksMy plan

Demo Account

Upcoming events

1 eventListMap

SEARCH

dasma

CATEGORY

All categories

CITY

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mm/dd/yyyy

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View details

Saved ✓

RSVP

Map view

Open a venue in Google Maps for directions.

CA

Campus Music Night

DLSU-D Activity Center - Dasmariñas